Humanitarian Crisis Neglect Index With Uncertainty Quantification and Counterfactual Reasoning

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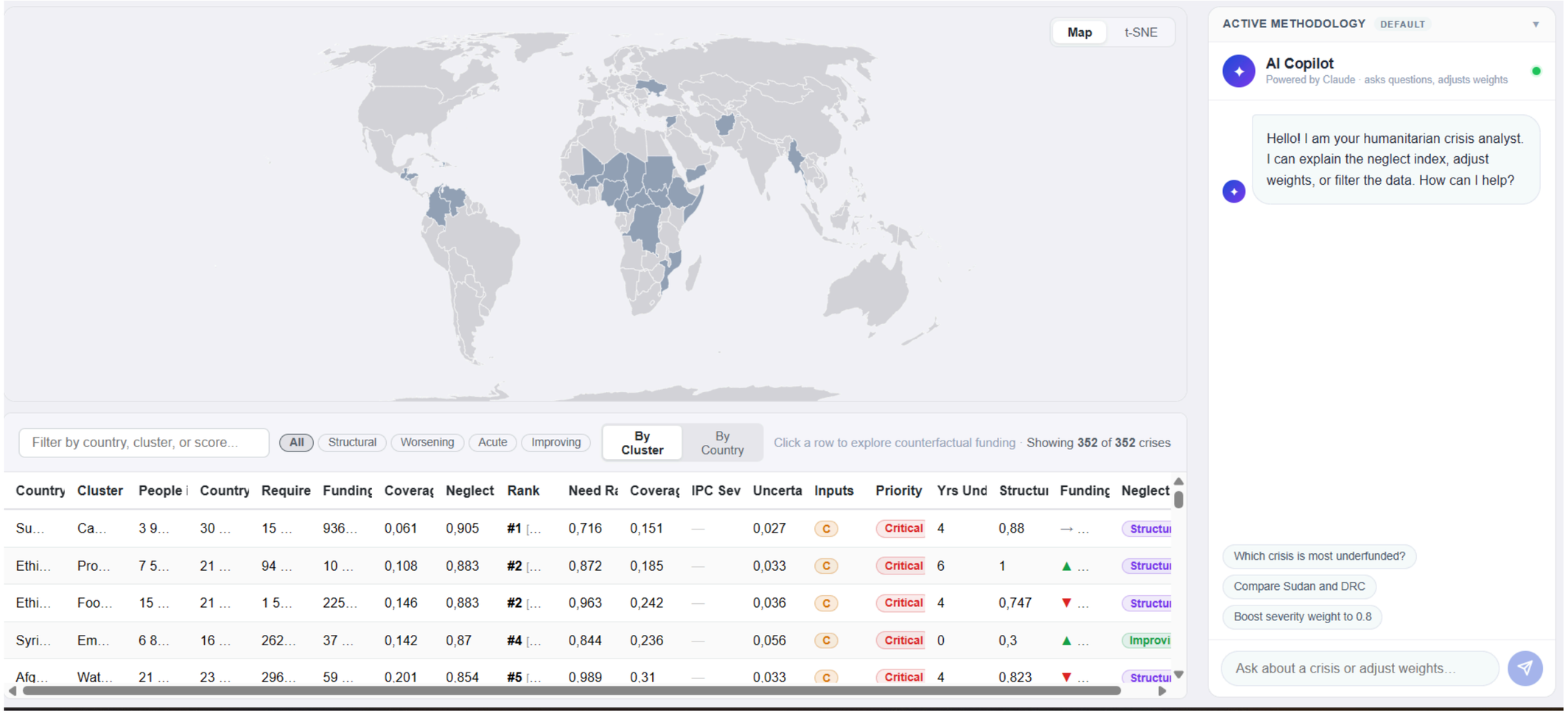
Our solution



Neglect index

$$\mathcal{N}(c) \;=\; \widetilde{S}_w \cdot \mathrm{severity}(c) \;+\; \widetilde{G}_w \cdot \big(1 - \mathrm{coverage_rank}(c)\big)$$

Severity

Case A (IPC + conflict events available)

$$\text{severity}(c) \;=\; \widetilde{N}_w \cdot \text{need_rank}(c) \;+\; \widetilde{I}_w \cdot \text{ipc_severity}(c) \;+\; \widetilde{E}_w \cdot \text{events_rank}(c)$$

Case B (IPC only)

$$\text{severity}(c) \;=\; \widetilde{N}_w \cdot \text{need_rank}(c) \;+\; \widetilde{I}_w \cdot \text{ipc_severity}(c)$$

Case C (conflict events only)

$$\text{severity}(c) \;=\; \widetilde{N}_w \cdot \text{need_rank}(c) \;+\; \widetilde{E}_w \cdot \text{events_rank}(c)$$

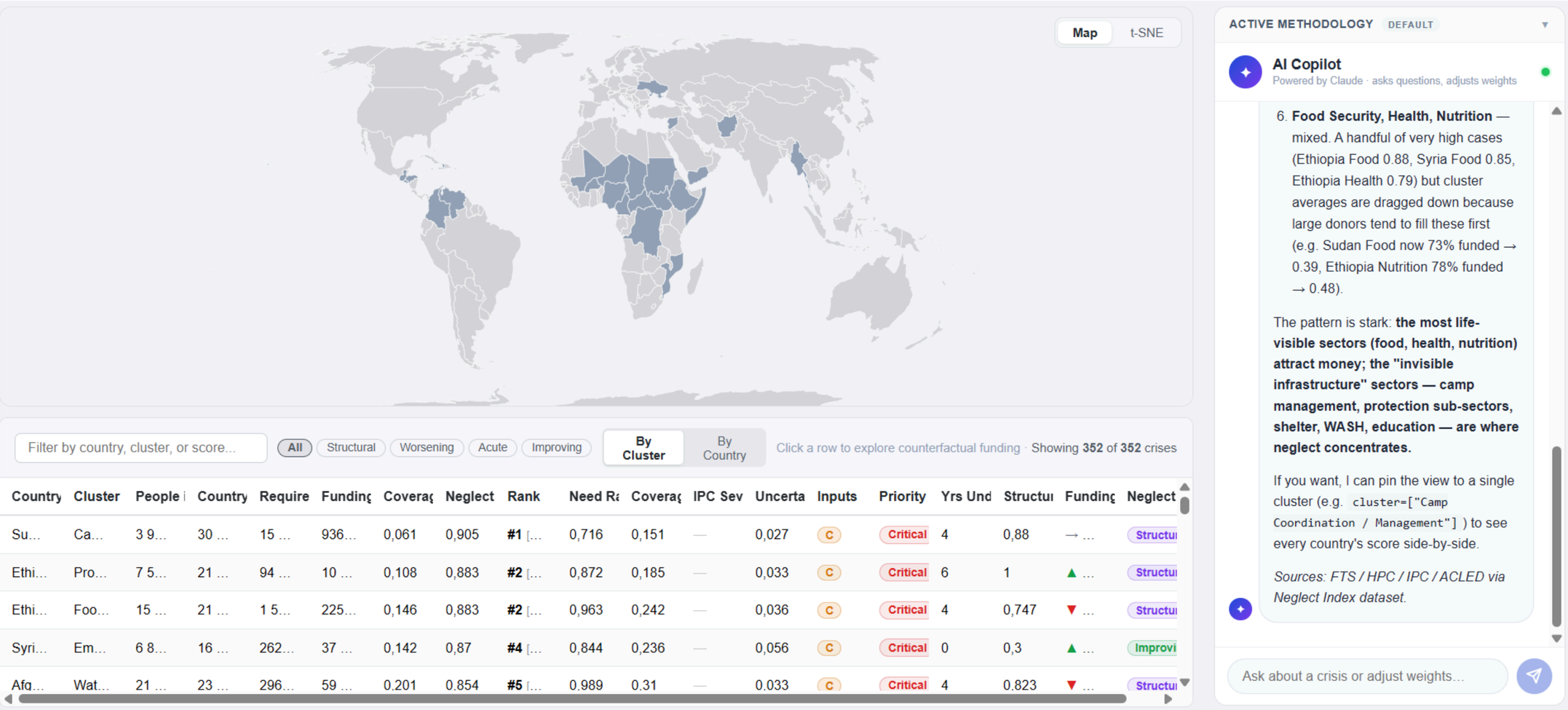
Case D (neither IPC nor events)

$$\text{severity}(c) \;=\; \text{need_rank}(c)$$

Severity Neglect Score

$$\text{sn_score} = 0.6 \cdot \frac{\min(\text{streak}, 5)}{5} + 0.4 \cdot \text{pct_years_underfunded}$$

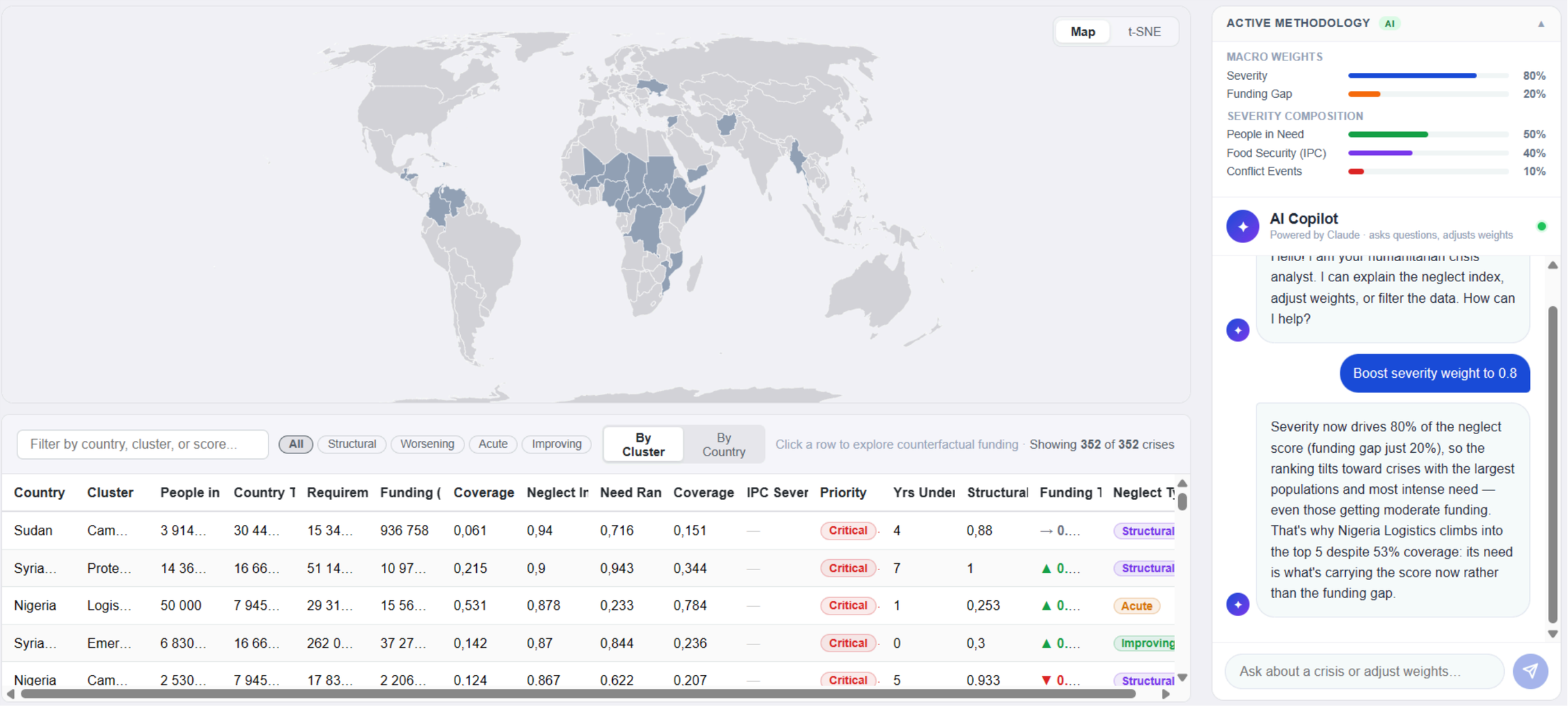
Clusters sorted by neglect index



Country	Cluster	People	Country	Require	Funding	Coverag	Neglect	Rank	Need Ra	Coverag	IPC Sev	Uncerta	Inputs	Priority	Yrs Und	Structur	Funding	Neglect
Su...	Ca...	3 9...	30 ...	15 ...	936...	0,061	0,905	#1 [...	0,716	0,151	—	0,027	C	Critical	4	0,88	→ ...	Structur
Ethi...	Pro...	7 5...	21 ...	94 ...	10 ...	0,108	0,883	#2 [...	0,872	0,185	—	0,033	C	Critical	6	1	▲ ...	Structur
Ethi...	Foo...	15 ...	21 ...	1 5...	225...	0,146	0,883	#2 [...	0,963	0,242	—	0,036	C	Critical	4	0,747	▼ ...	Structur
Syri...	Em...	6 8...	16 ...	262...	37 ...	0,142	0,87	#4 [...	0,844	0,236	—	0,056	C	Critical	0	0,3	▲ ...	Improvi
Afg...	Wat...	21 ...	23 ...	296...	59 ...	0.201	0.854	#5 [...	0.989	0.31	—	0.033	C	Critical	4	0.823	▼ ...	Structur

t-SNE analysis

AI Chatbot


Map
t-SNE
Filter by country, cluster, or score...
All
Structural
Worsening
Acute
Improving
By Cluster
By Country
Click a row to explore counterfactual funding · Showing 352 of 352 crises
Country Cluster People in Country T Requirem Funding (Coverage Neglect Ir Need Ran Coverage IPC Sever Priority Yrs Unde Structural Funding T Neglect T
Sudan Cam... 3 914... 30 44... 15 34... 936 758 0,061 0,94 0,716 0,151 — Critical 4 0,88 → 0.... Structural
Syria... Prote... 14 36... 16 66... 51 14... 10 97... 0,215 0,9 0,943 0,344 — Critical 7 1 ▲ 0.... Structural
Nigeria Logis... 50 000 7 945... 29 31... 15 56... 0,531 0,878 0,233 0,784 — Critical 1 0,253 ▲ 0.... Acute
Syria... Emer... 6 830... 16 66... 262 0... 37 27... 0,142 0,87 0,844 0,236 — Critical 0 0,3 ▲ 0.... Improving
Nigeria Cam... 2 530... 7 945... 17 83... 2 206... 0,124 0,867 0,622 0,207 — Critical 5 0,933 ▼ 0.... Structural
ACTIVE METHODOLOGY AI
MACRO WEIGHTS
Severity 80%
Funding Gap 20%
SEVERITY COMPOSITION
People in Need 50%
Food Security (IPC) 40%
Conflict Events 10%
AI Copilot
Powered by Claude · asks questions, adjusts weights
analyst. I can explain the neglect index, adjust weights, or filter the data. How can I help?
Boost severity weight to 0.8
Severity now drives 80% of the neglect score (funding gap just 20%), so the ranking tilts toward crises with the largest populations and most intense need — even those getting moderate funding. That's why Nigeria Logistics climbs into the top 5 despite 53% coverage: its need is what's carrying the score now rather than the funding gap.
Ask about a crisis or adjust weights...

Counterfactual Tool

Counterfactual Tool

Live HDX integration

“What would happen if the Yemen Food Security cluster received an extra $200 million?”

An extra $200M would be significant but wouldn't fundamentally change Yemen's status as a neglected food security crisis. Here's why:

The arithmetic. Yemen's Food Security appeal is **$1.10 billion**, currently funded at **$238M (21.6% coverage)**. An extra $200M would push funding

Live HDX integration

Can you extend the data with HDX data? I want a more in-depth understanding

The live HDX data adds depth to Yemen's Food Security story in three important ways — and one of the findings actually changes the picture I gave earlier.

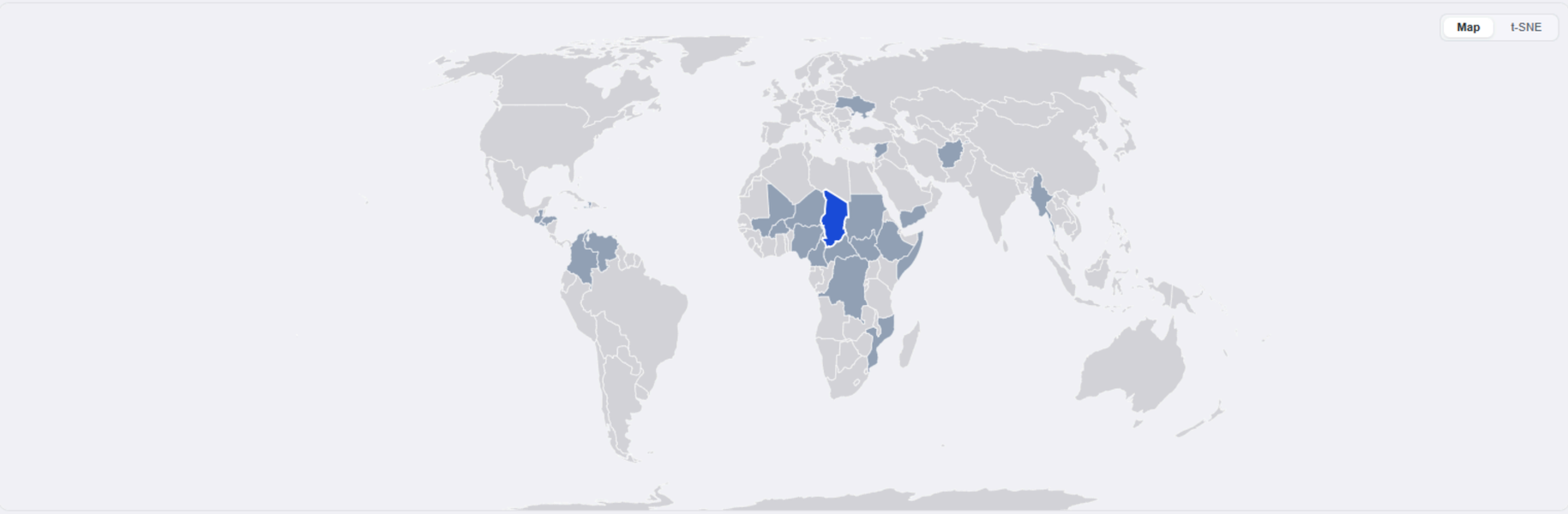
1. The funding collapse is even sharper than the CSV suggests

The full FTS history of Yemen's Humanitarian Response Plan shows one of the clearest "donor exit" patterns in the humanitarian system:

Year	Appeal (USD)	Funded
2018	$3.1B	**80%**
2019	$4.2B	**87%** ← peak

Sources: HDX HAPI — FTS Funding (appeals 2010–2026), IPC Food Security Phases (Yemen, 2018–2026 projections), ACLED Conflict Events (civilian targeting), Operational Presence (2025 cluster mapping). Ibb governorate used as a representative sample for sub-national IPC and presence data; cluster-level funding figures from the pre-computed Neglect Index dataset.

Confidence intervals



TCD

All | Structural | Worsening | Acute | Improving | By Cluster | By Country

Click a row to explore counterfactual funding · Showing 13 of 352 crises

Country	Cluster	People in N	Country Tot	Requiremer	Funding (US	Coverage	Neglect Indi	Rank	Need Rank	Coverage R	IPC Severity	Uncertainty	Inputs	Priority	Yrs Underfu	Structural S	Funding Tre	Neglect Type
Chad	Health	2 837 5...	7 049 226	40 069 ...	12 654 ...	0,316	0.446	#225 [17...	0,642	0,526	—	medium 0.04	C	Medium	2	0,573	▲ 0.015	Acute
Chad	Coordin...	0	7 049 226	26 845 ...	18 075 ...	0,673	0.415	#250 [19...	0,108	0,869	—	medium 0.04	C	Medium	1	0,32	▲ 0.046	Acute
Chad	Protecti...	300 017	7 049 226	6 355 035	1 065 9...	0,168	0.350	#292 [26...	0,267	0,27	—	medium 0.03	C	Low	2	0,573	▲ 0.048	Acute
Chad	Camp ...	653 811	7 049 226	7 223 718	1 989 736	0,275	0.340	#302 [21...	0,33	0,455	—	high 0.073	C	Low	2	0,573	▲ 0.075	Acute
Chad	Education	1 594 8...	7 049 226	32 603 ...	14 933 ...	0,458	0.317	#308 [24...	0,509	0,736	—	high 0.070	C	Low	0	0,2	▼ 0.014	Adequate
Chad	Water S...	3 488 2...	7 049 226	42 695 ...	23 284 ...	0,545	0.250	#327 [31...	0,693	0,796	—	medium 0.04	C	Low	1	0,32	▲ 0.082	Acute
Chad	Emerge...	710 297,5	7 049 226	34 044 ...	11 641 ...	0,342	0.183	#340 [32...	0,347	0,568	—	medium 0.04	C	Low	2	0,507	▲ 0.048	Acute

Thank you
for your attention!